Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/22/2024
PI Name:	Shippen, Dorothy Ph.D.		
Project Title:	Telomere Dynamics and Oxidative Stress in Arabidopsis in the Space Radiation Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology SPACE BIOLOGY--Radiation biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dshippen@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-862-2342
Organization Name:	Texas A&M AgriLife Research		
PI Address 1:	Department of Biochemistry and Biophysics		
PI Address 2:	300 Olsen Blvd, Suite 103, 2128 TAMU		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843	**Congressional District:**	10
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBPS E.9: Plant Studies
Start Date:	11/21/2022	**End Date:**	11/20/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA KSC
Contact Monitor:	O'Rourke, Aubrie	**Contact Phone:**	
Contact Email:	aubrie.e.orourke@nasa.gov		
Flight Program:			
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Bailey, Susan Ph.D. (Colorado State University) Wyatt, Sarah Ph.D. (Ohio University)		
Grant/Contract No.:	80NSSC23K0302		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose an International Space Station (ISS) Flight Research Investigation using Spectrum hardware, and radiation experiments at NASA Space Radiation Lab (NSRL, Brookhaven) and Colorado State University. This proposal leverages the expertise of three international leaders in the fields of telomere, radiation, and plant biology to explore the relationship between the space radiation environment, oxidative stress, and the dynamics of plant telomeres. Our preliminary data show that, in response to spaceflight, telomerase enzyme activity in Arabidopsis is strongly induced, but telomere length is steady. Telomerase activity was not substantially elevated by simulated microgravity conditions, implying that space radiation is responsible for telomerase induction. Our central hypothesis is that exposure to the space radiation environment results in elevated Reactive Oxygen Species (ROS) which in turn leads to genome oxidation and the high-telomerase activity phenotype observed in the telomere maintenance machinery of space flown plants. We will test this hypothesis with three specific Objectives. Objective 1 will examine the effect of the combined stressors associated with spaceflight on plant telomere dynamics, genome oxidation, and organellar stability in Arabidopsis seedlings grown aboard the ISS as compared to ground controls. We will employ a series of mutant and overexpression lines that influence telomeres and genome oxidation. Objective 2 will explore the impact of space radiation exposure on these parameters using the same Arabidopsis lines by studying the effect of fractionated schemes of simulated space radiation exposure (Galactic Cosmic Ray Simulator at the NSRL) as well as low dose, low dose-rate neutron and low dose gamma-ray exposure (irradiation facilities at Colorado State University). In Objective 3, we synthesize the data gathered from spaceflight and on-ground fractioned radiation exposure experiments to address our central hypothesis. This work is highly impactful as telomeres are an important biological marker of survivability in both plants and humans. Elucidating the relationship between the space radiation environment and telomere dynamics may prove to be a critical component of long-duration missions in the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research addresses how plants endure the impact of the space radiation environment. Our experiments may provide new insight into how different stressors alter the plant genome and the oxidative stress response, and the type of genetic modifications that can be made to enhance survival in harsh environments, including those that currently exist on Earth.
Task Progress:	PI: Dorothy E. Shippen, PhD, Texas A&M University, College Station, TX Co-PI: Sarah E. Wyatt, PhD, Ohio University, Athens, OH Co-PI: Susan M. Bailey, PhD, Colorado State University, Fort Collins, CO TITLE: Telomere dynamics and oxidative stress in Arabidopsis in the space radiation environment Period covered by report: Sept 23, 2023 to Sept. 22, 2024 Summary of Year 2 Research Accomplishments Pre-Flight, Ground Experiments In year 2 we focused on Ground Verification Test (GVT), preparation for Experiment Verification Test (EVT) and testing the impact of radiation (at NASA Space Radiation Laboratory/NSRL) on telomeres, telomerase, and genome oxidation status. During the middle of year 2 it was determined that Spectrum would not be a viable hardware for this project due to ongoing uncertainty regarding the timing of essential software updates. The experiment was shifted to the NASA Vegetable Production System (Veggie) hardware. This swap will not only achieve the original scientific objectives, but also enables more science through analysis of additional genetic lines with no compromise in data quality. We obtained all the genetic lines necessary for GVT/EVT at NASA Kennedy Space Center (KSC), on the International Space Station (Station) and for ground experiments at NSRL and Colorado State University. PF.3 – Optimize seedling growth: With the switch to Veggie, we were able to follow a similar protocol to the one successfully used by the Wyatt lab for Advanced Plant Experiment (APEX-07). This approach employed polyethersulfone (PES) membranes on agar in 10 cm square Petri dishes for growth of 12-day-old seedlings. Because younger seedlings are preferred for our analysis (telomerase is more enzymatically active at this developmental stage), we asked if 7-day-old seedlings would provide sufficient biomass/plate for our analysis. This was indeed the case. PF.4 – Optimize DNA and protein extraction: We completed this set of experiments for all the genetic lines and verified that sufficient yield and quality of nucleic acid and proteins could be obtained by growth of these genotypes. GVT: PI Wyatt accompanied B. Barcenilla and E. Canaday to KSC in July 2024 for GVT. Seedlings were grown for 7 days in Veggie and KSC staff sent materials back to Texas A&M for analysis. Seedling growth was assessed and found to be excellent. DNA and protein were extracted, and the yield and quality were measured. In the initial experiments there were a few samples with low DNA purity and yield. The suboptimal quality appeared to be caused by the phenol chloroform reagent we used. Our standard (preferred) phenol chloroform reagent from Thermo Fisher Scientific was not available at that time and so we substituted a reagent purchased from Sigma for extraction of the nucleic acids. When we obtained phenol chloroform from Thermo Fisher Scientific and re-extracted nucleic acids from the GVT samples, the quality and quantity of DNA improved substantially so that all samples were in the acceptable range. Thus, the Thermo Fisher reagent will be exclusively used for all future experiments. Ground Radiation Experiments (Objective 2) 2.1 – Radiation experiments at NSRL: To emulate space radiation exposures to in-flight Arabidopsis seedlings (wild type and mutant lines), B. Barcenilla and E. Canaday traveled to NSRL in May 2024 for experiments using Galactic Cosmic Ray Simulation (GCRSim). Seedlings were grown on plates for 7 days at NSRL and plates were exposed to three acute radiation doses via the GCRSim corresponding to 7 days onboard ISS (~4mGy total dose); 6 weeks on the International Space Station (ISS) (simulate seed to seed growth) (~25mGy total dose) and 6 months on the ISS (~10cGy total dose). After radiation delivery, seedlings were frozen as soon as possible (within 30 min.) to minimize DNA repair, and samples were shipped to the Shippen lab for processing and analyses. As expected, we found no change in telomere length. However, there was a statistically significant increase in genome oxidation, but no change in telomerase enzyme activity. Experiments are underway to examine these parameters in the mutants. However, the absence of telomerase induction in wild type plants was surprising. Additional time prior to sample freezing might be needed to allow a full response. Alternatively, the radiation dose at NSRL may not accurately mimic flight exposure (e.g., South Atlantic Anomaly). Finally, combined stressors may be needed to induce telomerase to the same extent as spaceflight; radiation plus microgravity, for example. We took advantage of the remarkable radiation facilities at NSRL to irradiate a large variety of seeds with 500 mGy

	gamma irradiation (dose corresponding to 1 year on the lunar surface). These seeds will not only provide an exceptionally valuable resource to examine the impact of radiation on telomeres in plants derived from the seeds, but will also enable us to study combined effects of radiation and other types of stress, including growth in lunar regolith simulant (see below). Additional experiments and new collaborations: Shippen lab personnel funded by this grant are also studying how plant growth in lunar regolith simulant impacts telomeres and telomerase. We discovered profound and progressive telomere shortening, indicating that genome stability is severely compromised in this substrate. We published the initial results (Barcenilla et al., 2024) and are collaborating with scientists in the Soil and Crop Sciences department (Texas A&M University / TAMU) to determine if this phenomenon occurs in crop plants. In collaboration with TAMU Biophysicist Dmitry Kurouski, we found that Ramen Spectroscopy can be used as a novel non-invasive method to assess stress in space-flown plants (Rodriguez et al., submitted).
Bibliography Type:	Description: (Last Updated: 06/03/2026)
Articles in Peer-reviewed Journals	Barcenilla BB, Meyers AD, Castillo-González C, Young P, Min JH, Song J, Phadke C, Land E, Canaday E, Perera IY, Bailey SM, Aquilano R, Wyatt SE, Shippen DE. "Arabidopsis telomerase takes off by uncoupling enzyme activity from telomere length maintenance in space." Nat Commun. 2023 Nov 29;14:7854. https://doi.org/10.1038/s41467-023-41510-4 ; PMID: 38030615; PMCID: PMC10686995 , Nov-2023
Articles in Peer-reviewed Journals	Barcenilla BB, Kundel I, Hall E, Hilty N, Ulianich P, Cook J, Turley J, Yerram M, Min J-H, Castillo-González C, Shippen DE. "Telomere dynamics and oxidative stress in Arabidopsis grown in lunar regolith simulant." Front Plant Sci. 2024 Feb 15;15:1351613. https://doi.org/10.3389/fpls.2024.1351613 ; PMID: 38434436; PMCID: PMC10908177 , Feb-2024